

Urban Expansion Analysis

Constantine, Algeria (2015-2020)

Full Methodology & Step-by-Step Workflow Documentation

Project	Portfolio Project 1 - GIS / Urban Growth Analysis
Software	QGIS (Desktop GIS)
Data source	European Commission, Joint Research Centre (JRC) - GHSL
Dataset	GHS-BUILT-S R2023A, 100m resolution
Boundary source	OpenStreetMap (Nominatim / Overpass relation export)
Study period	2015 - 2020 (5-year interval, both measured epochs)
Coordinate systems	WGS84 (EPSG:4326) for boundary, World Mollweide (ESRI:54009) for GHSL rasters

1. Project Objective

The goal of this project is to measure and visualize how the built-up (urbanized) area of Constantine, Algeria, changed between 2015 and 2020, and to identify where that growth was spatially concentrated. Rather than describing the current state of the city, this project performs a real **change detection** analysis between two measured time points, which is a standard method in remote sensing and urban planning studies.

Research question: *How did the built-up area of Constantine change between 2015 and 2020, and where was urban expansion concentrated?*

A key methodological principle followed throughout the project: no percentage or area figure was stated unless it was actually calculated from the data, and it is made explicit that GHSL measures 'built-up surface' according to its own technical definition, not the city in an administrative or fully comprehensive land-use sense.

2. Data Sources

2.1 Administrative Boundary

The boundary of Wilaya de Constantine was obtained from OpenStreetMap. The relation (administrative boundary, level 4) was located via the Nominatim search on openstreetmap.org, then its OSM Relation ID was used on the companion service polygons.openstreetmap.fr, which converts an OSM relation into a clean downloadable polygon. The original (unsimplified) geometry was exported as GeoJSON and saved as `constantine_boundary.geojson`.

2.2 Built-up Surface Data (GHSL)

Built-up area data came from the **GHS-BUILT-S R2023A** product, produced by the European Commission's Joint Research Centre (JRC) as part of the Global Human Settlement Layer (GHSL) programme, freely distributed via the Copernicus Emergency Management portal (human-settlement.emergency.copernicus.eu). This raster product measures built-up surface (in square meters per 100m grid cell) and is available for fixed epochs from 1975 to 2030 in 5-year steps.

Important methodological decision: GHSL provides projected (modeled) values for 2025 and 2030 rather than values measured from actual satellite observations. To keep the analysis defensible, only the two most recent **observed** epochs were used: **2015** and **2020**. This was confirmed before downloading any data.

The correct spatial tile covering Constantine was identified as: R5_C19 (Mollweide global tiling grid), and verified visually by overlaying it with the boundary layer before committing to the download. Two files were downloaded:

- GHS_BUILT_S_E2015_GLOBE_R2023A_54009_100_V1_0_R5_C19.tif
- GHS_BUILT_S_E2020_GLOBE_R2023A_54009_100_V1_0_R5_C19.tif

3. QGIS Project Setup

A new, clean QGIS project was created (urban_expansion_constantine.qgz) and immediately saved to a dedicated project folder before adding any data, to avoid an 'Untitled Project' state and prevent accidental data loss.

Step-by-step:

- Project → New (empty project)
- Project → Save As → saved as urban_expansion_constantine.qgz inside a dedicated project folder

4. Loading the Data into QGIS

4.1 Boundary (vector layer)

- Layer → Add Layer → Add Vector Layer
- Source file: constantine_boundary.geojson
- Layer displayed correctly in EPSG:4326 (WGS84), matching the real shape of the wilaya

4.2 Built-up rasters

- Layer → Add Layer → Add Raster Layer
- Both GHS-BUILT-S GeoTIFF files (2015 and 2020) were added
- Correct coverage of Constantine was verified visually by zooming to the boundary layer extent and checking that the built-up raster values actually fall inside and around it (coordinates ~36.36N, 6.86E)

5. Clipping the Rasters to the Study Area

Both raster layers were clipped to the exact boundary of Constantine so that the analysis and area statistics only reflect the wilaya, not the full downloaded tile.

Tool used: Raster → Extraction → Clip Raster by Mask Layer

Input layer	Mask layer	Output
GHS_BUILT_S_E2015...tif	constantine_boundary	built_2015_clipped.tif
GHS_BUILT_S_E2020...tif	constantine_boundary	built_2020_clipped.tif

Option enabled: 'Match the extent of the clipped raster to the extent of the mask layer', which guarantees that both clipped rasters share the same spatial extent - a requirement for a valid pixel-by-pixel comparison in the next step.

6. Change Detection (Raster Calculator)

The core analytical step of the project: computing, for every 100m pixel, the difference in built-up surface between 2020 and 2015.

Tool used: Raster → Raster Calculator

Expression:

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"built_2020_clipped@1" - "built_2015_clipped@1"
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Output layer: urban_change_2015_2020.tif. Interpretation: a positive pixel value means built-up surface increased in that 100m cell between 2015 and 2020 (new urban growth); a value of 0 means no measured change.

7. Extracting the Final Statistics

Tool used: Processing Toolbox → Raster Analysis → Raster Layer Statistics, run on urban_change_2015_2020.tif.

Statistic	Value
Pixel count	224,880
Minimum	0 m ² (no decrease recorded anywhere)
Maximum	5,467 m ² (single largest pixel increase)
Sum	2,975,661 m ²
Mean	13.23 m ² / pixel

Conversion: $2,975,661 \text{ m}^2 \div 1,000,000 = \sim\mathbf{2.98 \text{ km}^2}$ of new built-up surface across the wilaya of Constantine between 2015 and 2020, with no measured decrease in built-up surface anywhere in the study area.

8. Cartographic Design

Initial attempts stacked all three rasters (Built-up 2015, Built-up 2020, Urban Change) as overlapping layers in a single map. This produced a confusing, color-mixed result because the top layer visually hid the others. The layout was rebuilt using the standard approach for change-detection studies: **three separate, equally sized maps placed side by side** (before / after / change), rather than superimposed layers.

8.1 Symbology per layer

Layer	Color ramp	Purpose
Built-up Area - 2015	YlOrBr (yellow-orange-brown)	Baseline built-up surface

Built-up Area - 2020	Blues	Current built-up surface
Urban Change - 2015-2020	Reds	Net increase in built-up surface

8.2 Visual refinements applied

- All three maps use the identical extent and scale, so they are directly comparable side by side.
- Pixels with value 0 on the change layer were made transparent (Layer Properties → Transparency → custom transparent pixel range 0-0 at 0% opacity), so only actual change is visible instead of a solid colored background.
- The color stretch maximum on the change layer was manually lowered from the raw maximum (5,467) to a smaller value (~500) purely for display purposes. This is a visualization choice only - it does not alter the underlying data or the statistics reported above - and was made because a linear stretch across the full 1-5,467 range made most real change pixels appear nearly white.
- Layer names were renamed from raw file names (e.g. built_2015_clipped) to readable labels (e.g. 'Built-up Area - 2015') so the legend reads cleanly.
- The legend was cleaned to remove redundant 'Band 1 (Gray)' sub-labels and duplicate entries, keeping the numeric scale only on the Urban Change layer, since that is the layer that needs a quantitative key.

9. Final Map Layout & Export

Built with: Project → New Print Layout

Layout elements added:

- Title label: 'Urban Expansion - Constantine, Algeria (2015-2020)'
- Three equally sized map frames (Add Map), one per layer, created via copy-paste to guarantee identical size and extent
- A sub-title label above each map
- One shared scale bar (0 / 7.5 / 15 km), since all three maps share the same scale
- One north arrow
- A cleaned legend restricted to the Urban Change layer
- A data-source credit line at the bottom of the page

Data source line used on the final map:

Data source: European Commission, Joint Research Centre (JRC), GHS-BUILT-S R2023A (100m resolution) | CRS: World Mollweide (ESRI:54009) | Method: Raster difference (Built-up 2020 - Built-up 2015)

Exported via Layout → Export as PDF, at 300 dpi, as Urban_Expansion_Constantine_2015_2020.pdf.

10. Summary of QGIS Tools Used

Task	QGIS Tool / Menu
Load vector boundary	Layer → Add Layer → Add Vector Layer
Load raster data	Layer → Add Layer → Add Raster Layer
Clip raster to boundary	Raster → Extraction → Clip Raster by Mask Layer
Compute pixel difference	Raster → Raster Calculator
Compute area statistics	Processing Toolbox → Raster Layer Statistics
Style rasters	Layer Properties → Symbology (Singleband pseudocolor)
Make zero-value pixels transparent	Layer Properties → Transparency
Build the final map sheet	Project → New Print Layout
Export final map	Layout → Export as PDF

11. Known Limitations

- GHS-BUILT-S measures built-up **surface** per grid cell, not the administrative city boundary or all urbanized land uses (e.g. it does not directly represent roads, informal settlements below detection resolution, or non-structural land conversion).

- 100m spatial resolution means small or scattered individual buildings may not be resolved as distinct features.
- 2025 and 2030 GHSL epochs are model projections, not observations, and were deliberately excluded from this analysis for that reason.
- The color stretch on the Urban Change map (max ~500) is a display adjustment only; the reported statistics (sum, min, max) were computed from the unmodified raster values.

This document was written as the methodology record for Project 1 of a GIS portfolio (Urban Expansion Analysis series), intended to accompany the final map and the project's GitHub repository.